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```
clc; clearvars; close all;
```

General

```
syms t x var0 X var(t)
```

```
P = (1 / sqrt(2 * pi * var(t))) * exp( -((x - X)^2) / (2*var(t)));
```

Constants (for later)

```
XFixed = 5;  
x0 = 3;  
var0Fixed = 20;
```

Take Derivatives

```
% Temporal  
f1 = simplify(diff(P, t));  
fprintf("The temporal derivative is: \n\n");  
pretty(f1);  
  
% Spacial  
f2 = simplify(diff(diff(P, x), x));  
fprintf("The spacial derivative is: \n\n");  
pretty(f2);  
  
% Set the correct relation for var(t) and var0  
f1 = subs(f1, [X, var(t)], [XFixed, var0*t]);  
f2 = subs(f2, [X, var(t)], [XFixed, var0*t]);
```

The temporal derivative is:

$$\frac{\sqrt{2} \exp\left(-\frac{(X-x)^2}{2 \operatorname{var}(t)}\right)}{4 \sqrt{\pi} \operatorname{var}(t)^{5/2}} \frac{d}{dt} \left(-X^2 + 2 X x - x^2 + \operatorname{var}(t) \right)$$

The spacial derivative is:

$$\frac{\sqrt{2} \exp\left(-\frac{(X-x)^2}{2 \operatorname{var}(t)}\right)}{2 \sqrt{\pi} \operatorname{var}(t)^{5/2}}$$

Ratio

```
% Find the ratio alpha between f2 and f1
alpha = simplify(f2 ./ f1);
alpha = subs(alpha, x, x0);
fprintf("The diffusion constant \u03B1 between the derivatives is: \n\n");
pretty(alpha);
```

The diffusion constant α between the derivatives is:

$$\frac{2}{\operatorname{var}0}$$

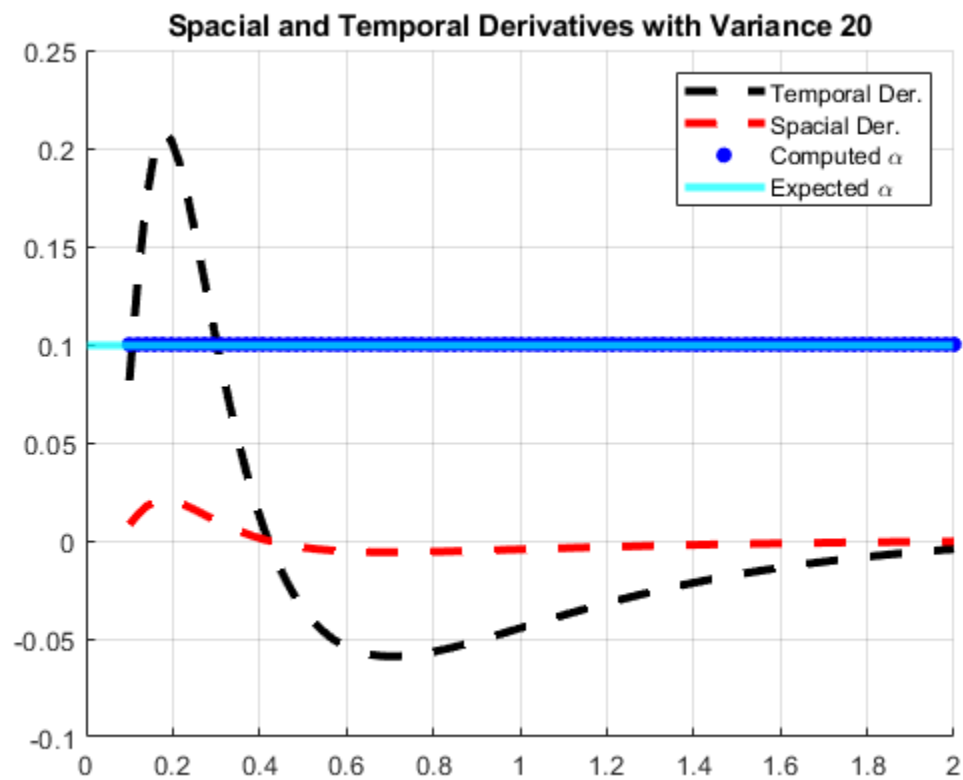
Plotting

```
tPlot = linspace(0.1, 2, 100);

% Substitute fixed values in so that it can be plotted
f1Plot = subs(f1, [var0, x], [var0Fixed, 5*t]);
f1Plot = double(subs(f1Plot, t, tPlot));

f2Plot = subs(f2, [var0, x], [var0Fixed, 5*t]);
f2Plot = double(subs(f2Plot, t, tPlot));

% Plot
lw = 3;
figure();
hold on
plot(tPlot, f1Plot, '--k', 'DisplayName', 'Temporal Der.', 'LineWidth', lw)
plot(tPlot, f2Plot, '--r', 'DisplayName', 'Spacial Der.', 'LineWidth', lw)
plot(tPlot, f2Plot./f1Plot, '.b', 'DisplayName', 'Computed \alpha',
'MarkerSize', 20)
yline(2/var0Fixed, 'c', 'DisplayName', 'Expected \alpha', 'LineWidth', lw)
grid on
hold off
legend()
title("Spacial and Temporal Derivatives with Variance " + string(var0Fixed));
```



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